

Basic Information

Experiment No.: 06

Date: 24-09-2026

Title: Java Deque Interface using ArrayDeque

Objective / Aim

To implement the Deque interface using "ArrayDeque" and demonstrate insertion, deletion, and accessing elements from both ends of the deque.

System Requirements

Hardware

- Computer/Laptop
- Minimum 4 GB RAM

Software

- Java JDK
 - Any Java IDE or Command Prompt
 - Operating System: Windows/Linux/macOS
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Algorithm / Logic Description

1. Create a "Deque" using "ArrayDeque".
2. Insert elements at the front using "addFirst()".
3. Insert elements at the rear using "addLast()".
4. Insert elements using "offerFirst()" and "offerLast()".
5. View the first and last elements using "peekFirst()" and "peekLast()".

6. Remove elements using "pollFirst()" and "pollLast()".
7. Remove elements using "removeFirst()" and "removeLast()".
8. Display the deque after each operation.
9. End the program.

UML Class Diagram

<<interface>>

Deque

1

1

implements

1

ArrayDeque

 $\wedge$

1

DequeExample

1

```
main()
```

1

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addFirst() addLast() offerFirst()

offerLast() peekFirst() peekLast()

pollFirst() pollLast() removeFirst()

1

removeLast()

Source Code

```
package collectionframework;

import java.util.Deque;
import java.util.ArrayDeque;

public class DequeExample {
    public static void main(String[] args) {

        Deque<String> dq = new ArrayDeque<>();

        // addFirst()
        dq.addFirst("B");

        // addLast()
        dq.addLast("C");

        dq.addFirst("A");
        dq.addLast("D");

        System.out.println("Deque: " + dq);

        // offerFirst()
        dq.offerFirst("Start");

        // offerLast()
```

```
    dq.offerLast("End");

    System.out.println("After offer: " + dq);

    // peekFirst()
    System.out.println("First: " + dq.peekFirst());

    // peekLast()
    System.out.println("Last: " + dq.peekLast());

    // pollFirst()
    System.out.println("Removed First: " + dq.pollFirst());

    // pollLast()
    System.out.println("Removed Last: " + dq.pollLast());

    System.out.println("Deque after removal: " + dq);

    // removeFirst()
    dq.removeFirst();

    // removeLast()
    dq.removeLast();

    System.out.println("Final Deque: " + dq);
}
}
```

Compilation & Execution Commands

Save the file as:

DequeExample.java

Since the program contains:

package collectionframework;

the file should be inside the "collectionframework" folder.

Compile:

```
javac DequeExample.java
```

Run:

```
java collectionframework.DequeExample
```

Output Screen

Deque: [A, B, C, D]

After offer: [Start, A, B, C, D, End]

First: Start

Last: End

Removed First: Start

Removed Last: End

Deque after removal: [A, B, C, D]

Final Deque: [B, C]

Result

The Deque interface was successfully implemented using "ArrayDeque", and insertion, deletion, and accessing operations from both the front and rear were successfully demonstrated.